Owen Yeo

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I am a product-focused Software Engineer specialising in full-stack AI orchestration, high-impact system design, and technical solutioning. Proven track record across the stack—from building low-latency computer vision pipelines to architecting multi-platform agentic workflows for enterprise operations. I am adept at bridging the gap between deep technical implementation and stakeholder-driven business value, be it in Mandarin or English. I am also a quick learner, learning skills from CUDA programming to building sequential search recommendation models on the job. Looking forward to working on meaningful projects and learning as much as I can!

EDUCATION

National University of Singapore **Aug 2022 – Present**

Bachelor of Computing, Major in Computer Science with specialisations in AI/ML and SWE

WORK EXPERIENCE

Software Development Intern, Care Corner (Singapore) **May 2026 - Present**

- Architected an end-to-end AI chatbot ecosystem using LangChain and LangGraph for complex orchestration, automating the initial triage of incoming clients, eliminating manual spreadsheet-based screening, and providing a first touchpoint for potentially distressed clients.
- Translated non-technical business requirements into scalable automation solutions, architecting data pipelines that streamlined internal operations and eliminated 20 hours of manual processing weekly.
- Engineered compliance and safety guardrails for multi-platform deployment (Telegram/TikTok), implementing age-verification protocols to ensure strict data privacy alignment with regional age-of-consent laws (such as PDPA standards for minors under 15).
- Containerised and deployed telemetry systems via Docker onto Google Cloud Platform (GCP) to establish live observability and seamless maintenance of deployed AI products.

Algorithms Engineer Intern, Shopee (Singapore) **Mar 2025 - Jul 2025**

- Evaluated and integrated advanced sequential recommendation algorithms (HSTU, ReaRec, LUM) using user interaction history, driving a 15% improvement in retrieval accuracy over the baseline DIN model.
- Executed large-scale feature engineering and data augmentation workflows within the e-commerce retrieval and ranking pipelines using Hadoop.
- Maintained production-grade software quality across the full SDLC, utilizing Kubernetes, Docker, and Prometheus for containerized deployment and system monitoring.
- Assisted in research on model improvements using Swing Neighbours in Prompt Engineering to improve model retrieval results. Observed a 10% improvement in model inference MAP.

Machine Learning Engineer Intern, JUSTRA AI (Guangzhou) **Jul 2024 – Jan 2025**

- Achieved a 75% computational speed-up across real-time inference pipelines by writing custom CUDA kernels, implementing hardware acceleration, and executing model quantisation via TensorRT.
- Architected a multimodal computer vision pipeline utilising two deep learning models—analysing hyperspectral data for plastic classification and RGB feeds for spatial coordinate mapping—to drive an automated air-spray sorting mechanism.
- Designed and documented a modular camera interfacing SDK using software design patterns (e.g., Singleton classes, layered architecture) to ensure seamless cross-team deployment across English- and Mandarin-speaking testing teams.
- Spearheaded the end-to-end development of a physical prototype by translating stakeholder requirements, negotiating with external vendors, and autonomously managing delivery milestones to meet critical deadlines.

PROJECTS

PeerPrep [https://github.com/CS3219-AY2526Sem1/cs3219-ay2526s1-project-g11]

- Architected a scalable microservices backend using Elixir and Go, enabling highly concurrent, real-time collaborative code editing with robust fault tolerance using CDRT.
- Orchestrated containerised deployments on Google Cloud Platform (GCP) utilizing Kubernetes and Docker, integrating load balancing and logging for comprehensive system observability.
- Established automated CI/CD pipelines to streamline testing and deployment, maintaining strict release quality and zero-downtime deployments across staging and production environments.

DriftDetector [https://driftx-w9rwtnecp5bmidct8lukeg.streamlit.app/]

- Engineered a lightweight MLOps monitoring platform featuring a modular FastAPI backend to wrap production inference APIs and capture critical request telemetry, including latency and throughput.
- Implemented statistical drift metrics (Population Stability Index and KL-Divergence) mapped to a real-time Streamlit dashboard, providing instant visibility into feature-level distribution shifts and model health.
- Containerised the entire application ecosystem using Docker to guarantee environment consistency and accelerate deployment cycles.

VirtualVault and CovAI [https://new-stockpicker.vercel.app]

- Developed scalable Node.js backend APIs to manage secure authentication, payment processing gateways, and dynamic routing for a full-stack MERN application.
- Authored CovAI, an LLM-powered CLI tool that parses local code coverage reports and automatically generates targeted unit and integration tests to push codebases past strict coverage thresholds.
- Integrated a responsive React frontend with the backend services, applying software engineering best practices for modularization and API documentation to ensure long-term maintainability.

Jobbly [https://github.com/owenyeo/jobbly]

- Architected an autonomous, event-driven tracking system using Next.js and Node.js, implementing BullMQ and a Redis message queue to reliably decouple long-running web scraping cron jobs from the frontend rendering cycle.
- Engineered a stateful LLM evaluation pipeline with LangGraph and the DeepSeek API, extracting structured entities from raw DOM data and utilizing Supabase pgvector to calculate cosine similarity scores between job descriptions and technical profiles.
- Designed a multi-stage heuristic funnel and execution logging system that prevented unhandled process crashes during API timeouts, and reduced expensive LLM embedding calls by filtering out unqualified data points prior to vectorisation.

Hackathons/Events

- BitHacks 2023
- Project Sa'Bai 2023
- Hack'n'Roll 2024
- TikTok Techjam 2025
- Interest Group head for Code4Good at NUS College

Skills

- Programming Languages: JavaScript, TypeScript, C++, Python, Java, Elixir, Go, Bash, SQL.
- Frameworks/Libraries: PyTorch, React, Node.js, FastAPI, Express, Streamlit, LangChain, LangGraph, MCP,
- DevOps/Infra: Docker, Kubernetes, GCP, TensorRT, CUDA, Redis, Kafka, Hadoop, Prometheus, Power Automate, Vercel, AWS Lambda
- Non-Technical: English, Mandarin, Conversational Thai, Canva, Leadership, Event Planning, Project Direction
- Selected Coursework: CS4248 (NLP), CS4225 (Big Data), CS4218 (Software Testing), CS3210 (Parallel Programming), CS2107 (Computer Security), CS2105 (Computer Networks), CS3216 (Software Engineering Principles), CS3230 (Design and Analysis of Algorithms)
- Interests: Mountaineering, Rock Climbing, Running, Guitar, Cooking, Reading